PHY4605 Computational Methods in Physics
Learning Note | Week 3

# Linear Systems: Assemble and validate two-loop Kirchhoff equations

*Turn two familiar circuit equations into a small linear system that can be solved, checked, and interpreted.*

**Physical question.** Two resistor loops share one resistor. If the source voltages and resistances are known, how can we find the two loop currents without losing track of signs, units, or physical meaning?

## Learning outcomes

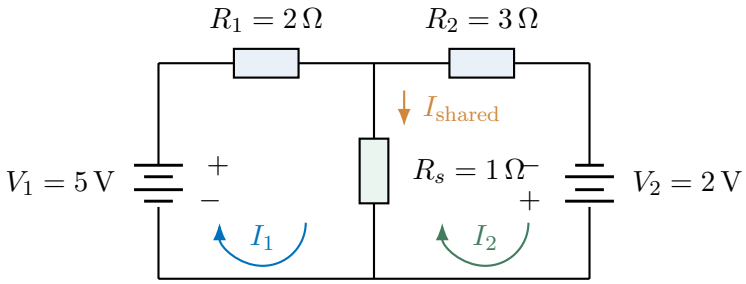
1. derive two consistent Kirchhoff equations from a stated current convention;
2. map those equations to $A\mathbf{x} = \mathbf{b}$ and solve them with MATLAB backslash; and
3. validate the solution by direct substitution and interpret the current signs physically.

***How to use this note.*** *Follow the numbered route from circuit to equations, code, validation, and physical interpretation. Pause at each checkpoint and predict before reading the result.*

**Capstone bridge.** Review the available capstone topics before the Week 04 preference checkpoint. Discuss one preferred and one alternate topic with your group; record the preference in Week 04. No capstone submission or grade is due in Week 03.

## 1. Start from the circuit, not the matrix

Choose both mesh currents clockwise. Let the left private resistor be $R_1 = 2\,\Omega$, the right private resistor $R_2 = 3\,\Omega$, and the shared resistor $R_s = 1\,\Omega$. The source rises are $V_1 = 5\,\mathrm{V}$ and $V_2 = 2\,\mathrm{V}$. The battery polarities below make those rises explicit: the long plate is positive, so the left source is traversed upward and the right source downward by the clockwise mesh currents.




Across the shared resistor, the two mesh currents oppose each other. At the top shared junction, $I_1$ enters from the left, while $I_2$ leaves to the right and $I_{\text{shared}}$ leaves downward through $R_s$. The junction rule (KCL) therefore says that current entering equals current leaving:

$$I_1 = I_2 + I_{\text{shared}},$$

so, if the shared-branch direction follows mesh 1,

$$I_{\text{shared}} = I_1 - I_2.$$

**Which Kirchhoff rule is used?** The *junction rule* (KCL) gives the shared-branch relation above. The *loop rule* (KVL) says that the algebraic sum of voltage changes around each closed loop is zero. We will use KCL at the shared junction, then KVL once around each loop to obtain two equations.

**Prediction checkpoint.** Both chosen mesh-current directions should give positive currents. The larger source and smaller private resistance suggest $I_1 > I_2$.

## 2. Write Kirchhoff equations step by step

Use the loop rule in the form *source rise minus resistor drops equals zero.* The circuit-to-equation mapping is shown term by term, using the clockwise directions and the KCL result above.

**Left loop.** The clockwise path goes upward through the left source, across the private resistor $R_1$, and downward through the shared resistor. Thus the source contributes a rise $+V_1$, the private resistor contributes a drop $R_1 I_1$, and the shared branch contributes a drop $R_s(I_1 - I_2)$:

$$\underbrace{+V_1}_{\text{source rise}} - \underbrace{R_1 I_1}_{\text{private-resistor drop}} - \underbrace{R_s(I_1 - I_2)}_{\text{shared-resistor drop}} = 0.$$

Rearranging gives

$$R_1 I_1 + R_s(I_1 - I_2) = V_1,$$

or

$$(R_1 + R_s) I_1 - R_s I_2 = V_1.$$

**Right loop.** The clockwise path goes downward through the right source, across the private resistor $R_2$, and upward through the shared resistor. In this upward traversal, the signed shared-branch current is $I_2 - I_1$, so the shared-resistor drop is $R_s(I_2 - I_1)$:

$$\underbrace{+V_2}_{\text{source rise}} - \underbrace{R_2 I_2}_{\text{private-resistor drop}} - \underbrace{R_s(I_2 - I_1)}_{\text{shared-resistor drop}} = 0.$$

Therefore

$$-R_s I_1 + (R_2 + R_s) I_2 = V_2.$$

Here $R_1$, $R_2$, and $R_s$ are resistance quantities; $\Omega$ is their unit, not the symbol for resistance. For example, $R_s I_{\text{shared}}$ has units $\Omega \times \text{A} = \text{V}$, so every term in each loop equation is a voltage.

## 3. Map each equation to one matrix row

The two equations become

$$\begin{bmatrix} R_1 + R_s & -R_s \\ -R_s & R_2 + R_s \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}.$$

With the values used in this example,

$$\underbrace{\begin{bmatrix} 3 & -1 \\ -1 & 4 \end{bmatrix}}_{A\ (\Omega)} \underbrace{\begin{bmatrix} I_1 \\ I_2 \end{bmatrix}}_{\mathbf{x}\ (\text{A})} = \underbrace{\begin{bmatrix} 5 \\ 2 \end{bmatrix}}_{\mathbf{b}\ (\text{V})}.$$

**Key idea.** A matrix is only a compact storage format for the two physical equations. Every coefficient must be traceable back to one Kirchhoff term.

## 4. Audit the matrix before solving

Before asking MATLAB for a solution, read the matrix back as physics. The safest habit is to point to one entry at a time and ask: *which unknown does this multiply, which loop equation does it belong to, and why does it have this sign?*

| Entry | Value | Physical origin |
|---|---|---|
| $A_{11}$ | $3\,\Omega$ | $R_1 + R_s$: resistance multiplying $I_1$ in the left-loop equation |
| $A_{12}$ | $-1\,\Omega$ | $-R_s$: effect of $I_2$ on the shared-resistor drop in the left loop |
| $A_{21}$ | $-1\,\Omega$ | $-R_s$: effect of $I_1$ on the shared-resistor drop in the right loop |
| $A_{22}$ | $4\,\Omega$ | $R_2 + R_s$: resistance multiplying $I_2$ in the right-loop equation |

The diagonal terms are positive sums because each mesh current passes through its own private resistor and the shared resistor. The off-diagonal terms are negative because the other mesh current opposes the chosen current through the shared resistor. The source vector is $\mathbf{b} = [5\ 2]^T\text{V}$. This sign pattern comes from the chosen arrows, not from a matrix rule to memorise.

**Units audit.** Row 1 reads $(3\,\Omega)I_1 + (-1\,\Omega)I_2 = 5\,\text{V}$. Both terms on the left are voltages. Row 2 has the same dimensional structure. If one coefficient were accidentally entered in volts or amperes instead of ohms, the matrix multiplication would no longer represent Kirchhoff's voltage law even if MATLAB still accepted the numbers.

### A quick coefficient-tracing routine

Before solving, ask three questions for any entry: *which physical equation is this row, which unknown is this column, and why does this coefficient have this sign and unit?* For example, $A_{12} = -1\,\Omega$ means that the coefficient multiplying $I_2$ in the left-loop equation is $-R_s$.

**Representation checkpoint.** Cover the equations above and use only the matrix to reconstruct the left-loop equation. Then cover the matrix and rebuild its first row from the left-loop equation. You should be able to move in both directions.

## 5. Pseudocode before MATLAB

1. store the resistances and source voltages;
2. build one matrix row from each Kirchhoff equation;
3. store the two source voltages in a column vector;
4. solve the two unknown currents;
5. reconstruct the shared-branch current;
6. substitute the currents back into both original equations; and
7. interpret the current signs and magnitudes.

## 6. Solve the supplied system in MATLAB

MATLAB's backslash operator solves the linear system $A\mathbf{x} = \mathbf{b}$; it is not ordinary division. Here, `A_ohm\b_V` asks MATLAB to find the currents that satisfy both equations.

Listing 1: Assemble the two-loop Kirchhoff system and solve for the mesh currents.

```
R1_ohm = 2; R2_ohm = 3; Rs_ohm = 1;
V1_V = 5; V2_V = 2;
A_ohm = [R1_ohm+Rs_ohm, -Rs_ohm; ...
         -Rs_ohm, R2_ohm+Rs_ohm];
b_V = [V1_V; V2_V];
x_A = A_ohm\b_V;
I1_A = x_A(1);
I2_A = x_A(2);
```

The solution is $I_1 = 2\,\text{A}$ and $I_2 = 1\,\text{A}$. The reconstructed shared-branch current is

$$I_{\text{shared}} = 2 - 1 = 1\,\text{A}.$$

## 7. Required validation: substitute into the original equations

For the left equation,

$$3(2) - 1(1) = 5\,\mathrm{V},$$

which reproduces the 5 V source. For the right equation,

$$-1(2) + 4(1) = 2\,\mathrm{V},$$

which reproduces the 2 V source.

Listing 2: Reconstruct each loop equation and test the voltage residuals.

```
left_V = (R1_ohm+Rs_ohm)*I1_A - Rs_ohm*I2_A;
right_V = -Rs_ohm*I1_A + (R2_ohm+Rs_ohm)*I2_A;
check_V = [left_V-V1_V; right_V-V2_V];
assert(max(abs(check_V)) < 1e-12)
```

**Validation result.** Both reconstructed loop voltages match the original sources. This directly checks the equations that represent the circuit.

## 8. Read the numerical solution back into the circuit

Both currents are positive, so the actual mesh currents follow the assumed clockwise arrows. A negative result would mean that the actual current runs opposite to the chosen arrow, not automatically that the calculation failed.

The shared-branch current is the difference because the mesh currents oppose one another:

$$I_{\text{shared}} = I_1 - I_2 = 1\,\mathrm{A}.$$

The contributions partially cancel; a negative difference would mean the actual shared-branch current is opposite to the assigned direction.

## 9. Diagnose a plausible matrix error

If the first off-diagonal coefficient is entered as $+R_s$ instead of $-R_s$, MATLAB can still return a result, but it need not satisfy the physical equations. Substitute it into the original Kirchhoff equations and inspect the mismatch.

**Responsible-AI caution.** An AI-generated matrix can be syntactically perfect yet physically wrong. Trace each coefficient to the stated direction and equation, then validate the solved currents independently.

| Common mistake | What may happen | Best first check |
|---|---|---|
| Wrong shared-resistor sign | MATLAB returns plausible but physically inconsistent currents | reconstruct the original KVL equations |
| Rows of $A$ swapped but $\mathbf{b}$ not swapped | each equation is paired with the wrong source | read each row together with its matching source entry |
| Unknown order changed | current values are assigned to the wrong labels | state the column order of $\mathbf{x}$ before solving |
| Units mixed | numerical scale is physically meaningless | audit coefficient $\times$ unknown = source units |

### Guided extension: equation ordering

Read the supplied reordered system: swapping rows is harmless only when the same swap is applied to $A$ and $\mathbf{b}$. This changes listing order, not the physical system.

### Guided extension: sensitivity intuition

Read the supplied $V_2 = 2.1\,\mathrm{V}$ comparison and interpret why both currents change through the shared resistor. Formal conditioning is not required.

### Optional extension: residual, rank, conditioning, and power

If useful, read the supplied diagnostics $A\mathbf{x} - \mathbf{b}$, 'rank(A)', and 'cond(A)', then interpret source power versus resistor dissipation. These distinction-level checks are optional, not required for Week 3.

## Three takeaways

1. Choose unknown directions and derive the Kirchhoff equations before constructing a matrix.
2. Each row of $A\mathbf{x} = \mathbf{b}$ must preserve both the physical sign convention and the units.
3. A solver result becomes credible only after direct substitution and physical interpretation.

**Preparation for the practical.** Be ready to transfer the same workflow to another two-loop circuit and to a two-node KCL model. For each case, state the unknowns and units, map the equations to $A\mathbf{x} = \mathbf{b}$, solve with backslash, and validate the original equations before trusting the result.